Efficient Template Distinction Modeling Tracker With Temporal Contexts for Aerial Tracking

Hong Zhang, Wanli Xing, Huakao Lin, Hanyang Liu, and Yifan Yang

***Abstract*— In recent years, trackers based on neural networks have demonstrated excellent tracking performance. Compared to general tracking tasks, aerial tracking tasks require more stringent operational efficiency of the tracker. Furthermore, since aerial devices such as unmanned aerial vehicles usually shoot from a high altitude, the tracked target occupies fewer pixels, resulting in scarce target discrimination information and more susceptibility to interference from cluttered backgrounds. Unfortunately, existing trackers usually model the entire template region nondifferently, which tends to confuse target and background information in the template and reduce the robustness of the tracker in complex scenes. Additionally, most tracking networks refuse or only refer to a single feature layer to learn the risky temporal context information, which makes it difficult to accurately supplement the scarce target discrimination information. To this end, we propose an efficient template distinction modeling tracker with temporal contexts, called ETDMT, designed to improve the complex scene robustness of aerial tracking through a combination of template distinction modeling and temporal context analysis. The tracker employs a template distinction modeling transformer network, which distinguishes between target and background elements and adopts different modeling approaches for different elements to alleviate the background interference problem prevalent in aerial tracking. Then, the temporal contexts of the tracker are complemented by a global–local spatial awareness update module, which enhances the tracker's understanding of the latest target state by comprehensively evaluating and adjusting dynamic templates. Extensive experiments demonstrate that the proposed ETDMT achieves advanced aerial tracking performance and efficient running speed.**

***Index Terms*— Aerial tracking, template distinction modeling, temporal context, transformer network.**

I. Introduction

DUE to the high maneuverability and safety characteristics of aerial devices, aerial tracking technology [1], [2], [3], [4] is playing an increasingly important role in the field of aerial remote sensing. It is widely used in scenarios such as traffic management, geographic detection, and disaster rescue. Aerial tracking tasks predict the location and scale of subsequent targets based on the initial bounding box given in the first frame shot by the aerial devices. However, unlike general tracking, aerial tracking usually faces unique challenges, including scarce target discrimination information due to aerial remote imaging, rapid changes in perspective brought by the high maneuverability of aerial devices, and the need for an efficient and lightweight design of aerial trackers.

Currently, aerial trackers are mainly divided into two major categories: correlation filter trackers and neural network trackers. As a representative of correlation filter trackers, minimum output sum of squared error (MOSSE) [5] first achieves excellent running speed and tracking performance. Then, kernelized correlation filter (KCF) [6] extends the multichannel filter to further improve the ability to express target features. Subsequently, to address the negative impact of boundary effects, spatially regularized discriminative correlation filters (SRDCF) [7] and background-aware correlation filters (BACF) [8], respectively, used fixed spatial constraints and binary matrix cropping for mitigation. Aberrance repressed correlation filter (ARCF) [9] suppresses the target aberrations caused by environmental interference during the detection process, enhancing the tracker's ability to represent the foreground. Some trackers [10], [11] employ the spatiotemporal regularization terms to improve the robustness of the tracker scenarios. These correlation filter methods achieve high running speed through cyclic sampling and frequency domain acceleration and well suit the speed needs of aerial tracking. However, the further improvement of tracking accuracy is limited by the relatively single feature extraction and fusion methods of correlation filter methods. By applying neural networks to aerial tracking tasks, some trackers [1], [12], [13], [14], [15] have achieved good tracking performance. SiamFC++ [12] achieves noticeable improvements in both running speed and tracking performance using efficient feature extraction networks and anchor-free regression networks. SiamAPN [1] constructs a new anchor proposal network for adaptive anchor proposing. SiamAPN++ [16] improves tracker performance by modeling the semantic relationships between individual feature maps and among different feature maps. TCTrack [17] and TCTrack++ [15] explore temporal context information, improving tracking performance from feature extraction and refinement of similarity maps. Nevertheless, these trackers make it difficult to perform early exchange of features. Aba-ViTrack [18] introduces the Aba-ViT network, which improves tracking efficiency by adaptively discarding tokens based on the assessment of each token's halting probabilities. However, this tracker does not fully explore the temporal context information of the target.

II. Motivation

Although existing aerial trackers have achieved impressive tracking performance and speed, they still suffer from two major problems.

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1) These trackers usually do not clearly distinguish between the real target area and the background area in the rectangular template. They model each area of the templates equally, which confuse foreground and background information in the template and exacerbates the interference of cluttered backgrounds with tracked targets that lack discriminative information.
2) Most trackers lack a multidimensional update evaluation mechanism. They do not update or only update according to the local region similarity, which leads to a weakened perception of background disturbances by the update module and easily introduces background interference into the template.

To address the first problem, we propose the template distinction modeling transformer network. The network divides the template into real target areas and background areas and models different relationships for them. Specifically, only the real target area in the template is directly associated with the search area and generates tracking results. The background area in the template is not directly involved in the prediction of the tracking results, but it is used for modeling within the template to highlight the target areas. This modeling approach prevents the direct impact of background area in the template on tracking results. Regarding the second problem, we propose a global–local spatial awareness update module. This module evaluates both global background interference and local target similarities, leading to a more comprehensive and accurate learning of the target's temporal contexts. The contribution of this article can be summarized as follows.

1) An efficient aerial tracking architecture with template distinction modeling and temporal contexts is proposed. It efficiently models the template differently and captures target appearance changes based on global interference and local similarity.
2) A template distinction modeling transformer network is developed. It explicitly divides the template areas and constructs different modeling pathways to block the direct influence of background on the tracking results.
3) A global–local spatial awareness update module is designed. It evaluates global background interference and local target similarity in the current search area to update the tracker accurately.
4) Extensive experimental results demonstrate that the proposed tracker achieves competitive tracking performance and running speed compared to other advanced trackers.

III. Related Works

A. Spatial Modeling in the Tracker

Spatial modeling, as the core of trackers, has always been a research hotspot. Correlation filter trackers, through modeling the target's handcrafted features and accelerating computation in the Fourier domain, achieve excellent tracking speed and performance. However, the expressive power of handcrafted features is limited, making it difficult for correlation filter trackers to maintain stable tracking in complex scenes. In contrast to correlation filter trackers, Siamese tracking networks, leveraging massive offline training samples, have gained robust feature expression and fusion capabilities, gradually attracting researchers' attention. As a representative work of Siamese tracking networks, SiamFC [19] simplifies the tracking task into a matching task of sample pairs and uses efficient cross correlation operations to search for targets in the search area. Subsequently, SiamRPN [20] introduces the region proposal network into tracking tasks, applied to bounding box regression. DaSiamRPN [21] improves the target's foreground discrimination ability through data augmentation and negative sample training strategies. SiamRPN++ [22] and SiamDW [23] use a deeper backbone network for spatial modeling of templates and search areas. Although the deeper backbone network improves tracking performance, it significantly increases the model's parameters. Considering that the hyperparameters present in anchor box regression reduce the robustness of trackers, anchor-free Siamese trackers [12], [24], [25] not only have a simpler regression structure but also improves the accuracy of spatial feature regression of bounding boxes. Recently, some trackers [26], [27], [28] begin to adopt transformer structure. Thanks to the long-range modeling capabilities of the transformer network, the global spatial modeling capabilities of trackers are further enhanced. Among them, OSTrack [28] designs a one-stream tracking structure but models all information in the template indiscriminately, easily allowing background information to interfere with localization accuracy. Moreover, although these trackers have strong feature expression capabilities, the huge parameter size brought by their heavy networks makes it difficult to meet the lightweight design requirements of aerial tracking. Trackers designed for aerial tracking tasks [1], [15], [16], [17], [18], have adopted more lightweight network structures. Some aerial trackers [1], [12], [16] based on CNN structures, although achieving a good balance between tracking speed and accuracy, are limited in further enhancing the tracker's target feature fusion ability due to the lack of global modeling capabilities of CNNs. Other trackers [13], [15], [17], [29] utilize CNN structure for feature extraction and transformer structure for feature fusion. However, these aerial trackers do not explicitly distinguish between the real target and background areas in the template images, which makes it difficult to correctly identify real target information. Different from these trackers, the proposed tracker distinguishes between the target and background areas in templates and employs a one-stream transformer network with template distinction modeling. It not only blocks the direct influence of background information in the template on tracking results but also promotes early feature fusion.

B. Temporal Contexts in the Tracker

Compared to the low risks associated with using spatial information, the incorrect utilization of temporal context information can significantly increase the risk of tracking failure. Traditional correlation filter tracking methods [6], [7], [30] typically employ linear updating to iteratively refine the template. However, due to the more robust target semantic features extracted by deep networks, many trackers [20], [26], [28] rely solely on the initial target template for the entire tracking process, inevitably leading to the negative impact of the template becoming outdated. Thus, some deep learning trackers have developed mechanisms to learn the target's temporal context information. Dsiam [31] uses a rapid transformation learning model to learn online from historical

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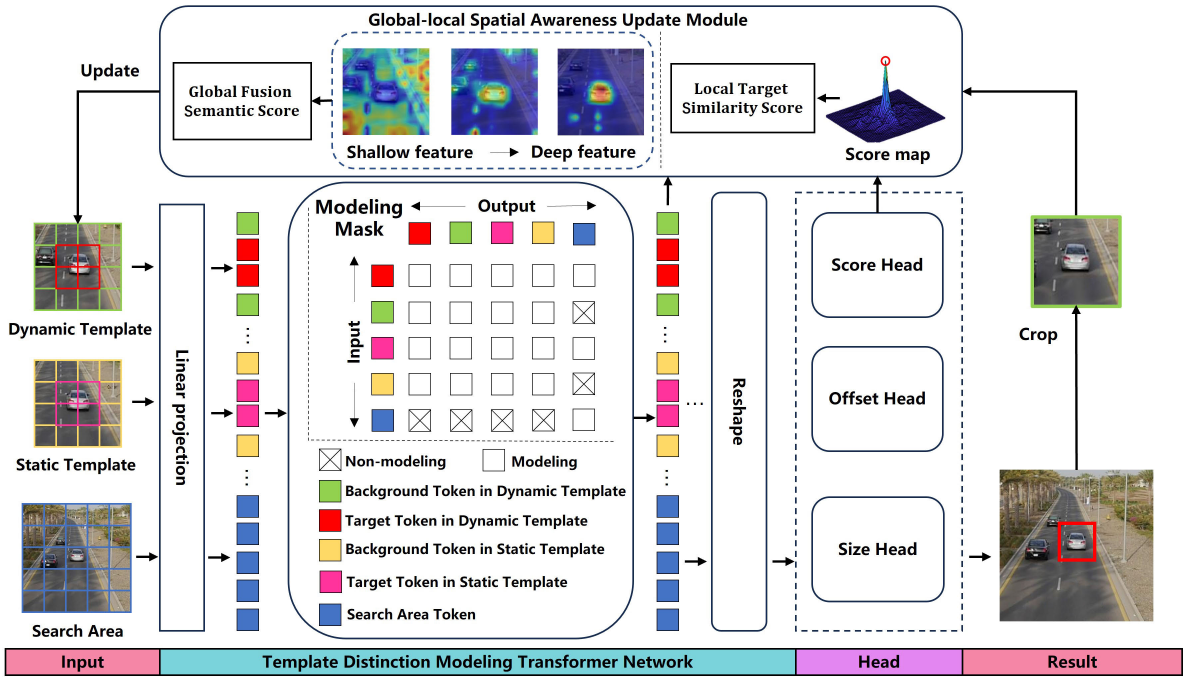


Fig. 1. Overall schematic of the ETDMT tracker.

data about changes in target appearance and background suppression. MemTrack [32] employs an LSTM structure to track targets based on historical samples. UpdateNet [33] combines collected samples to create new cumulative encoded templates. DiMP [34] maintains a specified number of historical samples and regularly activates an optimizer for updates. Ocean [24] has a DiMP-based update structure to learn the latest appearance information of the target continuously. Additionally, TCTrack [17] and TCTrack++ [15] have designed temporal feature extraction to learn temporal context information.

Recently, heavy transformer trackers [27], [35], [36] with temporal context information have mostly used explicit template update schemes to learn the latest target state. While these efficient update schemes are equally suitable for lightweight aerial trackers to efficiently acquire extensive contextual information about the target, it can be devastating for the tracker if errors occur, leading to failure. Therefore, the explicit template update schemes demand higher sample assessment accuracy. Most trackers [27], [35] typically adopt the deepest feature layer to evaluate update scores and control template updating. However, these trackers are too heavy to be applied to aerial platforms. In addition, this deepest feature tends to focus more on foreground expression, while undervaluing the background expression that is more contained in shallow features. To this end, the proposed tracker evaluates the global interference in the search area based on semantic features fused at different levels, and combines it with the local target similarity, to jointly assess and learn the temporal context information of the target from both a global and local perspective.

IV. Proposed Method

A. Overall

This section provides an overview of the proposed ETDMT tracker. As shown in Fig. 1, ETDMT mainly consists of two structures: the template distinction modeling transformer network and the global–local spatial awareness update module. Specifically, the template distinction modeling transformer network first divides the linearly mapped tokens into is divided into static template tokens, dynamic template tokens, and search area tokens. In addition, tokens from static templates are divided into the target token in the static template and the background token in the static template based on whether they belong to the real target area. Similarly, the dynamic template tokens are divided into the target token in the dynamic template and the background token in the dynamic template. Subsequently, along with the search area tokens, these tokens undergo the modeling masks to enable differential modeling of several areas in the templates, thus effectively blocking the direct negative impact of background interference in templates on tracking results. Then, the head network outputs the current frame's bounding box through the score head, the offset head, and the size head, and collects samples from the current frame. Finally, the collected sample of the current frame goes through the global–local spatial awareness update module to enable the tracker to accurately learn the target temporal contexts. In this update module, the global fusion semantic score combines shallow and deep features to assess the level of global interference in the search area and the local target similarity score uses the maximum peak in the score map to measure the predicted target similarity in the local region.

B. Template Distinction Modeling Transformer Network

1) Revisiting Original One-Stream Tracking Network: The input of the original one-stream tracking network [28] is a pair of images, consisting of a template $z \in \mathbb{R}^{3\times H_z\times W_z}$ and a search area $x \in \mathbb{R}^{3\times H_x\times W_x}$. They are divided and flattened into sequences in regions of 16×16, following the method of ViT [37]. The sequences are uplifted to a high-dimensional hidden space through linear mapping parameters and then

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combined with 1-D positional encoding to generate template tokens E_z and search area tokens E_x. Subsequently, these two types of tokens are sequentially concatenated to form the concatenated token $E_{zx} = [E_z; E_x]$, which is then fed into transformer encoder layers for relational modeling through attention operations Attn

$$\begin{aligned}\text{Attn}(Q, K, V) &= \text{SM}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) \times V \\ &= \text{SM}\left(\frac{[Q_z; Q_x] \times [K_z; K_x]^T}{\sqrt{d_k}}\right) \\ &\quad \times [V_z; V_x] \end{aligned} \tag{1}$$

where Q is the query, K denotes the key, and V represents the value. The subscripts z and x indicate that the element belongs to the template and the search area, respectively. SM is the softmax operation. Furthermore, attention modeling matrix AModel can be expressed in the following form:

$$\begin{aligned}\text{AModel} &= \text{SM}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) \\ &= \text{SM}\left(\frac{[Q_z\ {K_z}^T, Q_z\ {K_x}^T; Q_x {K_x}^T, Q_x {K_x}^T]}{\sqrt{d_k}}\right) \\ &= \begin{bmatrix} W_{zz}, W_{zx} \\ W_{xz}, W_{xx} \end{bmatrix} \end{aligned} \tag{2}$$

where W_{zz} and W_{xx} represent the self-modeling of the template and the search area, respectively. W_{zx} and W_{xz} denote the relevance modeling between the template and the search area.

2) Proposed Tracking Network: In tracking tasks, templates are typically set as rectangles, which inevitably leads to a large amount of background information within the rectangular template. Although the above-mentioned one-stream tracking network allows free interaction between the template and the search area, enhancing the tracker's feature expression capability, this method of free modeling between all template tokens and search area tokens can easily cause the tracker to struggle in distinguishing between the real target and background in the template, thereby reducing the tracker's environmental adaptability. Furthermore, these trackers usually employ a fixed input mode of the initial template and search area, limiting the tracker's ability to learn the target's temporal context information online.

To this end, we propose the template distinction modeling transformer network containing dynamic and static templates. The proposed network divides the template tokens into target tokens and background tokens based on the bounding box, where the tokens inside the bounding box are target tokens and the rest are background tokens, and establishes different relational models specifically for target tokens, background tokens, and search area tokens. Specifically, the input of the template distinction modeling transformer network comprises three images, represented as static template $z_s \in \mathbb{R}^{3\times H_s \times W_s}$, dynamic template $z_d \in \mathbb{R}^{3\times H_d \times W_d}$, and search area $x \in \mathbb{R}^{3\times H_x \times W_x}$. They similarly undergo segmentation, flattening, linear mapping, and 1-D positional encoding operations to generate dynamic template tokens E_d, static template tokens E_s, and search area tokens E_x. Subsequently, the dynamic template E_d is divided into tokens corresponding to the real target area E_{dt} and tokens corresponding to the background area E_{db}. Similarly, the static template E_s is divided into tokens corresponding to the real target area E_{st} and tokens corresponding to the background area E_{sb}. The search area tokens are not divided. The five types of tokens are sequentially concatenated to form the concatenated token $E_{sdx} = [E_{dt}; E_{db}; E_{st}; E_{sb}; E_x]$ for template distinction attention operation TDAttn

$$\begin{aligned}&\text{TDAttn}(Q, K, V) \\ &= \text{SM}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) \cdot \text{Mask} \times V \\ &= \text{SM}\left(\frac{[Q_{dt}; Q_{db}; Q_{st}; Q_{sb}; Q_x] \times [K_{dt}; K_{db}; K_{st}; K_{db}; K_x]^T}{\sqrt{d_k}}\right) \\ &\quad \cdot \text{Mask} \times [V_{dt}; V_{db}; V_{st}; V_{sb}; V_x] \end{aligned} \tag{3}$$

where the subscripts dt and db denote the real target area and the background area within the dynamic template, respectively. Similarly, the subscripts st and sb represent the real target area and the background area within the static template, while x represents the search area. Mask denotes the masking matrix. The template distinction attention modeling matrix TDModel can be expressed in the following form:

$$\begin{aligned}\text{TDModel} &= \text{SM}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) \cdot \text{Mask} \\ &= \begin{bmatrix} W_{dtdt} & W_{dtdb} & W_{dtst} & W_{dtsb} & 0 \\ W_{dbdt} & W_{dbdb} & W_{dbst} & W_{dbsb} & 0 \\ W_{stdt} & W_{stdb} & W_{stst} & W_{stsb} & 0 \\ W_{sbdt} & W_{sbdb} & W_{sbst} & W_{sbsb} & 0 \\ W_{xdt} & 0 & W_{xst} & 0 & W_{xx} \end{bmatrix} \\ \text{St. Mask} &= \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} \end{aligned} \tag{4}$$

where W_{dtdt}, W_{dbdb}, W_{stst}, W_{sbsb}, and W_{xx}, respectively, represent the self-modeling of the dynamic template's target area, dynamic template's background area, static template's target area, static template's background area, and the search area. W_{xdt} represents the modeling between the search area and the dynamic template's target area. The rest are similar. From (4), it is evident that the mask matrix efficiently eliminates the modeling results between the background areas of the two types of templates and the search area, retaining only the modeling between the target areas of the templates and the search area. This approach prevents the background areas in the templates from directly influencing the tracking results.

In summary, as illustrated in Fig. 2, there are two types of modeling for the static template: self-modeling and relevance modeling with the dynamic template. The dynamic template also employs two types of modeling: self-modeling and relevance modeling with the static template. The search area involves three types of modeling: self-modeling, relevance modeling with the target area of the dynamic template, and relevance modeling with the target area of the static template. Therefore, the core idea of template distinction modeling is

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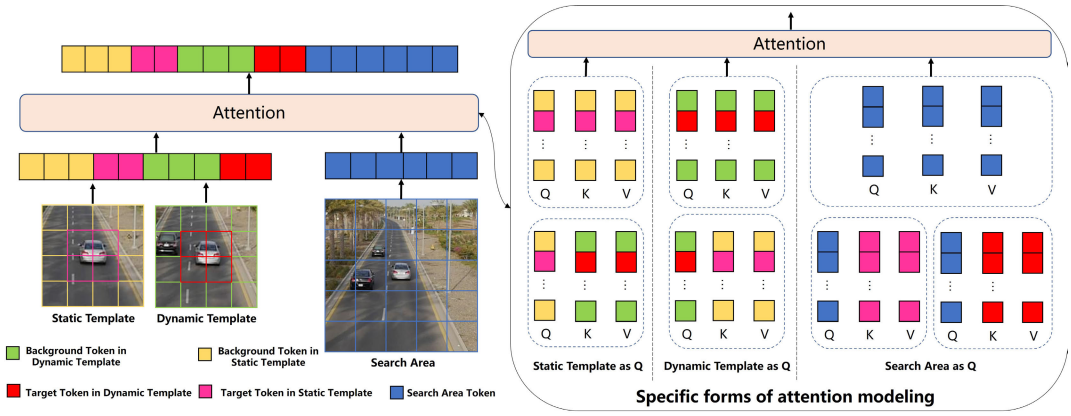


Fig. 2. Modeling forms in the template distinction modeling transformer network.

that only the target areas in these two templates are modeled directly with the search area to generate the bounding box. The background areas in the templates are used for the internal modeling of templates to highlight the target area and do not directly interact with the search area. This approach mitigates the decrease in tracking robustness caused by trackers confusing target and background information in the template.

C. Global–Local Spatial Awareness Update Module

While template update is an efficient way for trackers to learn the target temporal contexts, it is still very challenging to find samples that are correct and contain as little interference as possible. Existing transformer trackers often rely solely on the output from the deepest layer of ViT to generate update scores. This approach has limitations for sensing environmental interference and easily introduces interference in the tracker. To address this issue, this section introduces the global–local spatial awareness update module for adaptive evaluation and updating of the dynamic template. In this module, both the global fusion semantic score and the local target similarity score are applied to the dynamic template update. The global fusion semantic score evaluates the level of background interference globally in the search area, while the local target similarity score evaluates the target similarity of the predicted target region in the search area.

1) Global Fusion Semantic Score $Score_G$: After the dynamic template, static template, and search area are inputted into the template distinction modeling transformer network, the global–local spatial awareness update module extracts the search area tokens from three different layers of the template distinction modeling transformer network, from shallow to deep. These tokens are then rearranged in sequence into 2-D spatial feature maps M_1, M_2, and M_3. Subsequently, M_1, M_2, and M_3 are multiplied by weights ϕ_1, ϕ_2, and ϕ_3, respectively, and concatenated as $M_c = \text{Concat}[\phi_1 \cdot M_1, \phi_2 \cdot M_2, \phi_3 \cdot M_3]$. The weights ϕ_1, ϕ_2, and ϕ_3 correspond to the layer numbers of M_1, M_2, and M_3, respectively. Then, M_c is mapped as a single channel M_{c1} by three linear layers to mix semantic features from different layers progressively. Finally, M_{c1} passes through the 7×7 large kernel followed by the 3×3 small kernel and the 2×2 small kernel to generate the global fusion semantic score.

2) Local Target Similarity Score $Score_L$: The predicted target location in the current frame is the location with the highest score in the classification score map, where the classification scores stand for the target similarity of the corresponding region. Therefore, the proposed tracker ETDMT directly uses the highest classification score as the local target similarity score of the predicted region.

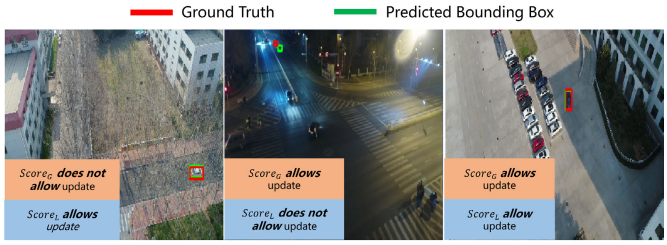


Fig. 3. Global score and local score work together for tracker update.

3) Update Method: Object tracking is applied to sequential images, which also means there is a significant redundancy of information between adjacent frames. Thus, updating frame-by-frame is sometimes not necessary. The proposed ETDMT tracker adopts an efficient group filtering strategy to manage the selection and updating of the dynamic template. Specifically, the tracker saves the collected samples of each frame and groups every adjacent N frame together, selecting the frame with the highest sum of Score_G and Score_L as the sample to be updated for the current group. When the sample to be updated satisfies $\text{Score}_G > 0.5$ and $\text{Score}_L > 0.5$, it will replace the dynamic sample to complete the tracker's update. The previously retained dynamic samples will be discarded after the update. As shown in Fig. 3, evaluating samples through global and local information reduces the interference in the samples to be updated and improves the correctness of the update. The overall structure of the global-local spatial awareness update module is shown in Fig. 4.

D. Head and Loss

The output search area tokens are reshaped into a 2-D spatial feature map and then fed into the head structure. The head structure includes a score head, an offset head, and a size head, all of which are composed of stacked convolutional layers, batch normalization layers, and ReLU activation functions. The score head outputs the classification score map (CMap), the offset head outputs the offset map (OMap), and the size head outputs the height and width of

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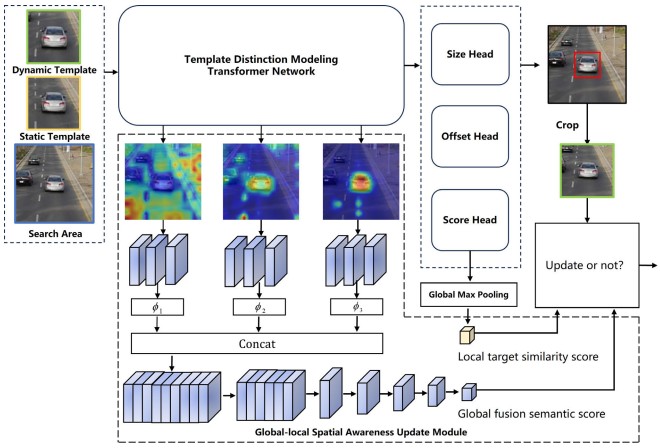


Fig. 4. Global–local spatial awareness update module.

the bounding box (SMap). The predicted target location is at the location of the highest classification score in CMap, and the target size is determined by the width and height at the corresponding location in SMap. The predicted target location and size (x, y, w, h) are calculated as follows:

$$\begin{aligned}(x_m, y_m) &= \text{argmax}(\text{CMap}) \\ (x, y) &= x_m + \text{OMap}(0, x_m, y_m), y_m + \text{OMap}(1, x_m, y_m) \\ (w, h) &= (\text{SMap}(0, x_m, y_m), \text{SMap}(1, x_d, y_d))\end{aligned} \tag{5}$$

where argmax() denotes the operation to locate the coordinates of the maximum value. (x_m, y_m) is the location of the maximum coordinate in CMap. CMap $\in [0, 1]^{2\times H_x/16\times W_x/16}$, OMap $\in [0, 1)^{2\times H_x/16\times W_x/16}$, and SMap $\in [0, 1]^{2\times H_x/16\times W_x/16}$. The tracker uses a weighted focal loss L_{wfl} [38] to train target classification, with labels generated using a Gaussian distribution. Complete intersection over union (IOU) loss L_{ciou} [39] and L_1 loss are adopted as the training frame for the bounding box. The overall loss function of the tracker is as follows:

$$L_{\text{all}} = L_{\text{wfl}} + \delta_{\text{ciou}} L_{\text{ciou}} + \delta_1 L_1. \tag{6}$$

Here, $\delta_{\text{ciou}} = 2$ and $\delta_1 = 5$ are the loss function coefficients, consistent with [27] and [28].

V. Experiments

A. Implementation Details

1) Model: The proposed template distinction modeling transformer network includes 12 repetitive attention modeling layers that perform both feature extraction and fusion capabilities. The input sizes are 128 × 128 pixels for the dynamic template, 128 × 128 pixels for the static template, and 256 × 256 pixels for the search area. Tokens from the search area at the 2nd, 7th, and 12th layers of the network are collected to calculate the global fusion semantic score. The tracker has three heads, each employing a lightweight network layer with four layers of stacked Conv-BN-ReLU.

2) Training: The training is conducted on two NVIDIA RTX3090 GPUs. The training datasets include LaSOT [40], COCO [41], TrackingNet [42], and GOT10K [43] (excluding 1K forbidden sequences from the GOT10K training set, following the convention [27]). Common data augmentation methods are used, such as horizontal flipping. The template distinction modeling transformer network takes ViT-Tiny [37] with CAEv2 [44] as initial parameters. The batch size for training is 32. AdamW is used as the optimizer. The initial weight for the backbone is set to 10^{-5}, and for other parts, it is 10^{-4}, with a weight decay of 10^{-4}. The training process is divided into two stages: the first stage consists of 260 epochs, primarily for training the backbone and head parameters, and the second stage includes 30 epochs, dedicated to training parameters in the global–local spatial awareness update module alone. Each epoch processes 60k pairs of images.

3) Testing: On an NVIDIA RTX2080Ti GPU, ETDMT runs at a speed of 127 frames/s. The number of adjacent frames N is set to 20. The commonly used Hanning window penalty in tracking tasks is applied as prior knowledge to adjust the classification results. In simple terms, a Hanning window of the same size is multiplied by the classification map, and the location with the highest score is selected as the predicted target location. Relevant experimental materials will be released at https://github.com/BULAB7/ETDMT/.

B. Evaluation Benchmarks and Metrics

In the testing experiment, four standard aerial tracking test datasets and their corresponding metrics are applied, including UAV123-10fps [45], DTB70 [46], UAVDT [47], and UAVTrack112-L [1].

1) Datasets: UAV123-10fps contains 123 sequences of large-scale aerial tracking benchmarks with a frame rate of 10 frames/s. DTB70 contains 70 tracking sequences, which mainly contain complex motion challenges with tracking challenges such as background clutter (BC) and scale variations (SVs). UAVDT consists of 100 video sequences that are captured in a variety of complex scenarios containing different flight altitudes, different camera fields of view, and a variety of weather conditions. UAVTrack112-L is a long-term aerial tracking benchmark with more than 60k frames. which contains tracking challenges such as low illumination.

2) Metrics: The tests use the commonly used success rate and precision as evaluation criteria. The success rate is calculated by the IOU between the predicted box and the bounding box, while precision is based on the center location error between the predicted bounding box and the ground truth. In addition, the tests also targeted an evaluation of 12 types of tracker attributes, including aspect ratio change (ARC), BC, SV, similar object (SOB), camera motion (CM), full occlusion (FOC), partial occlusion (POC), out of view (OV), viewpoint change (VC), low resolution (LR), fast motion (FM), and illumination variation (IV).

C. Overall Performance Comparison

First, our tracker is compared overall with 21 other excellent trackers, including TATrack-ViT [48], AVTrack-ViT [49], Aba-ViTrack [18], DRCI [50], SGDViT [29], ClimRT [51], TCTrack++(offline) [15], TCTrack [17], P-siamfc++ [52], F-siamfc++ [53], HiFT [13], SiamAPN++ [16], SiamAPN [1], AutoTrack [54], SiamFC++ [12], SiamDW [23], SiamRPN++ [22], ARCF [9], DaSiamRPN [21], ECO [55], and SiamFC [19].

1) Results on UAV123-10fps: The proposed ETDMT ranks first in success rate (66.6%) and precision (86.9%), as shown

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TABLE I
COMPARATIVE RESULTS OF FOUR TRACKING BENCHMARKS

Method	Source	UAV123-10fps		DTB70		UAVDT		UAVTrack112-L	
		Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.
ETDMT	Ours	66.6	86.9	67.1	85.8	60.7	84.1	64.2	81.5
TATrack-ViT [48]	TGRS2024	66.0	84.3	63.9	82.2	58.6	80.5	-	-
AVTrack-ViT [49]	ICML2024	65.7	83.2	63.3	81.3	57.7	79.9	-	-
Aba-ViTrack [18]	ICCV2023	65.5	85.0	66.4	85.9	59.9	83.4	64.2	81.1
DRCI [50]	ICME2023	55.2	73.6	61.8	81.4	59.0	84.0	-	-
SGDViT [29]	ICRA2023	58.5	76.6	60.3	80.6	-	-	-	-
ClimRT [51]	ICRA2023	57.5	76.0	-	-	-	-	-	-
TCTrack++(offline) [15]	TPAMI2023	59.9	78.0	62.6	81.5	-	-	-	-
TCTrack [17]	CVPR2022	58.8	77.4	62.2	81.3	53.0	72.5	58.2	78.6
P-siamfc++ [52]	ICME2022	54.9	73.1	60.4	80.3	56.6	80.7	53.1	70.4
F-siamfc++ [53]	IJCNN2022	54.5	72.1	60.5	81.4	56.6	80.7	54.5	74.2
HiFT [13]	ICCV2021	56.9	74.9	59.4	80.2	47.5	65.2	55.1	73.4
SiamAPN++ [16]	IROS2021	58.0	76.4	59.4	78.9	55.6	76.9	53.7	73.5
SiamAPN [1]	TGRS2021	56.6	75.2	58.6	78.4	51.7	71.1	-	-
AutoTrack [54]	CVPR2020	47.7	67.1	47.8	71.6	45.0	71.8	40.5	67.5
SiamFC++ [12]	AAAI2020	54.8	72.8	60.3	80.5	55.3	76.2	52.5	70.1
SiamDW [23]	CVPR2019	48.3	69.2	50.4	73.5	52.6	73.9	44.8	67.2
SiamRPN++ [22]	CVPR2019	55.1	73.5	58.9	79.5	-	-	-	-
ARCF [9]	ICCV2019	47.3	66.6	47.2	69.4	45.8	72.0	39.9	64.0
DaSiamRPN [21]	ECCV2018	48.3	69.2	47.2	69.4	-	-	-	-
ECO [55]	CVPR2017	46.8	64.0	44.8	63.5	41.6	64.9	41.7	64.8
SiamFC [19]	ECCV2016	47.3	68.0	48.3	71.9	-	-	-	-

in Table I. Compared to the ViT-based trackers Aba-ViTrack and SGDViT, ETDMT shows an improvement of 1.1% and 8.1% in success rate, and 1.9% and 10.3% in precision, respectively. Compared to the CNN-based tracker TCTrack, the ETDMT tracker shows improvements of 7.8% in success rate and 9.5% in precision. In addition, compared to the TATrack-ViT, ETDMT improves success rate and precision by 0.6% and 2.6%, respectively.

2) Results on DTB70: As shown in Table I, the proposed ETDMT ranks first in success rate (67.1%), 0.7% higher than the second-ranked tracker Aba-ViTrack. In precision, ETDMT (85.8%) is slightly lower by 0.1% than Aba-ViTrack, ranking second. Compared to SGDViT, ETDMT achieves a 6.8% improvement in success rate and a 5.2% improvement in precision. Moreover, ETDMT outperforms TATrack-ViT by 3.2% success rate and 3.6% precision.

3) Results on UAVDT: The proposed ETDMT achieves a success rate of 60.7% and a precision of 84.1%, ranking first in both metrics, as shown in Table I. Compared to Aba-ViTrack, ETDMT improves by 0.8% in success rate and 0.7% in precision. Additionally, compared to TCTrack, ETDMT shows improvements of 7.7% in success rate and 11.6% in precision. In addition, compared to TATrack-ViT, ETDMT further improves the success rate and precision by 2.1% and 3.6%, respectively.

4) Results on UAVTrack112-L: As shown in Table I, the proposed ETDMT achieves the same success rate as Aba-ViTrack but with higher precision. Compared to TCTrack, ETDMT exhibits superior tracking performance, improving by 6.0% in success rate and 2.9% in precision.

5) Results of Tracking Attributes: The proposed ETDMT is compared with seven advanced trackers for 12 tracking attributes on UAV123-10fps, including Aba-ViTrack [18], SGDViT [29], TCTrack++(offline) [15], TCTrack [17], SiamAPN++ [16], SiamAPN [1], and HiFT [13]. As shown in Table II, ETDMT achieves first place in precision across all 12 attributes. In the success rate comparison, ETDMT ranks first on 11 attributes and is only slightly below Aba-ViTrack on the OV attribute. Specifically, for the BC attribute, ETDMT has a precision of 72.0% and a success rate of 50.5%, which mitigates the negative effect of background interference on localization by suppressing the interactions between the background area in the template and the search area. Compared to the second-ranked Aba-ViTrack, it achieves an improvement of 5.6% in precision and 4.3% in success rate. Compared to TCTrack, ETDMT improves by 12.6% in precision and 10.3% in success rate. Additionally, in the IV attribute, ETDMT continuously learns illumination change information based on dynamic templates, thereby achieving a precision of 84.0% and a success rate of 62.9%.

D. Tracking Efficiency and Real-World Test

Regarding model parameters and computational volume, ETDMT has 6.7M parameters and 2.2G MACs, as shown in Table III. Compared to the advanced transformer tracker Aba-ViTtrack, ETDMT's parameters are down 1.3M and MACs are down 0.2G. In addition, the running results on an NVIDIA RTX2080Ti GPU show that ETDMT runs up to 62% faster than Aba-ViTtrack, reaching 127 frames/s, which proves the efficiency of the proposed ETDMT tracker.

Further, to verify the efficiency of ETDMT operating on the aerial platform, real-world tests are carried out on an onboard embedded processor. On the NVIDIA Jetson Xavier NX, an embedded on-board processor commonly used for aerial tracking tasks, the proposed tracker ETDMT achieves a real-time tracking speed of 40.7 frames/s without TensorRT acceleration, where the CPU and GPU utilization is 38% and 72%, respectively.

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TABLE II
ATTRIBUTE COMPARISON

Method	ARC		BC		SV		SOB		CM		FOC		POC		OV		VC		LR		FM		IV	
	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.
Ours	85.0	64.7	72.0	50.5	85.2	65.2	87.1	64.4	90.2	69.1	72.3	48.2	82.7	61.3	83.0	64.0	88.6	69.1	74.7	50.8	81.7	62.3	84.0	62.9
Aba-ViTrack	82.4	63.2	66.4	46.2	83.1	63.9	83.0	61.3	86.0	66.4	68.9	45.3	79.8	59.0	82.9	64.6	87.5	69.1	73.0	49.7	75.3	57.8	80.4	60.4
TCTrack++(offline)	75.1	56.5	59.6	41.0	76.2	58.2	77.0	56.7	75.0	57.5	61.7	39.4	71.3	52.1	70.8	53.6	75.8	60.0	63.7	42.6	67.4	49.8	70.8	52.5
TCTrack	73.0	55.0	59.4	40.2	74.7	56.7	74.0	53.0	74.5	56.9	57.6	35.5	71.2	51.5	70.4	53.8	74.0	58.9	63.0	41.1	67.3	49.8	69.9	51.5
SGDViT	73.1	54.9	53.6	36.1	74.5	56.7	74.0	54.2	74.4	57.3	55.4	34.4	68.1	49.5	67.3	51.5	74.5	59.2	61.7	41.0	68.9	51.3	67.9	51.0
SiamAPN++	71.7	53.8	54.5	37.9	73.5	55.7	71.3	52.1	76.1	58.5	57.0	35.9	68.4	50.1	71.7	55.2	75.3	58.9	59.6	39.7	71.1	52.2	69.1	51.3
SiamAPN	72.0	52.9	55.2	37.5	73.1	54.8	69.9	51.2	74.9	56.0	53.2	32.1	67.4	48.4	68.8	50.5	77.5	59.7	59.6	38.8	71.3	50.8	68.4	50.4
HiFT	69.6	51.8	53.3	36.5	72.5	54.9	67.3	49.0	73.2	56.1	52.5	32.3	65.8	47.9	68.5	51.4	71.4	56.1	55.1	36.5	68.4	49.7	66.8	48.8

TABLE III
SPEED COMPARISON

Method	Source	Speed	Param	MACs
HiFT	ICCV2021	174fps	10.0M	6.5G
LighTtrack	CVPR2021	91fps	2.0M	544.5M
TCTrack	CVPR2022	207fps	6.3M	4.5G
TCTrack++	TPAMI2023	196fps	7.1M	4.4G
Aba-ViTrack	ICCV2023	78fps	8.0M	2.4G
ETDMT	Ours	127fps	6.7M	2.2G

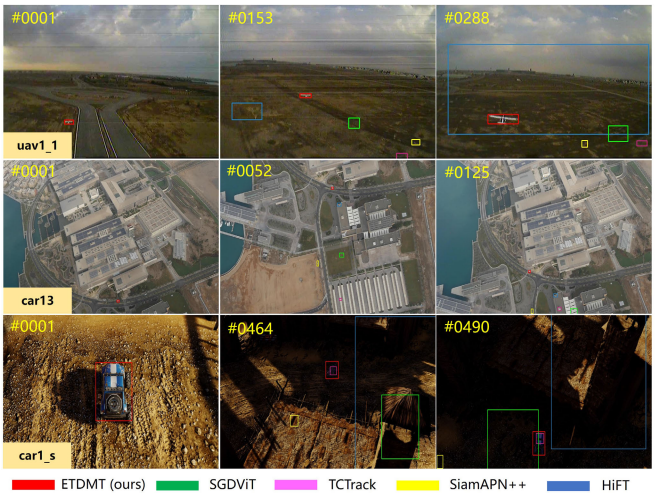


Fig. 5. Visualization sequences.

E. Visualization Sequences

As shown in Fig. 5, the proposed ETDMT tracker is visually compared with four other advanced trackers, including SGDViT [29], TCTrack [17], SiamAPN++ [16], and HiFT [13].

In the uav1_1 sequence, the tracking target is an aircraft. In this scenario, imaging noise, background clutter, and the target's rapid movement pose severe challenges for the trackers. At frame #153, the other four trackers, including SGDViT, TCTrack, SiamAPN++, and HiFT, experience tracking drift and fail to recover the target at frame #288. In contrast, ETDMT, with its template distinction modeling and temporal context information, reduces background interference and learns the target's appearance changes, thereby maintaining continuous tracking of the target.

In the car13 sequence, the tracking target is a distant vehicle. In this scenario, the tracker's robustness is tested by challenges like camera movement, small target imaging, and a complex environment. At frame #52, only ETDMT maintains normal tracking, while the other trackers drift due to the adverse environment and struggle to track the correct target subsequently. This demonstrates the robustness of our tracker in complex tracking environments.

In the car1_s sequence, the tracking target is a nearby car, subject to significant illumination changes. At frame #464, as the light intensity gradually decreases, only ETDMT and TCTrack can accurately locate the target, but ETDMT shows better scale estimation ability. At frame #490, as the light intensity decreases further, the ETDMT utilizes the temporal context information to quickly learn effective discriminative information under low-light conditions, resulting in stable scale prediction capabilities.

TABLE IV
TEMPLATE DISTINCTION RATIO ABLATION

Template distinction ratio	0.5	1.0	1.5	2.0
Prec.	82.3	86.3	84.6	83.5
Succ.	62.5	66.0	65.2	64.2

TABLE V
UPDATE INTERVAL ABLATION

Update interval N	5	10	15	20	25
Prec.	84.8	85.1	86.0	86.9	86.3
Succ.	64.7	65.1	65.9	66.6	65.9

F. Ablation Experiments

In this section, ablation experiments are conducted to verify the effectiveness of each component in the proposed ETDMT method. First, the parameter ablation experiment of the template distinction ratio and update interval N are performed on UAV123-10fps. The template distinction ratio in the proposed template distinction modeling transformer network is the ratio of selected real target areas to bound box areas. The update interval N in the proposed global–local spatial awareness update module refers to the number of adjacent frames in an update group. From Table IV, it can be seen that the tracker has better precision and success rates when the template distinction ratio is 1.0, which means that the area inside the bounding box is the real target area. When the ratio of the selected real target area to the tracking frame is less than 1.0 (e.g., a ratio of 0.5), the real target information that interacts with the search area will be significantly missing, resulting in a decrease in the discriminative ability of the tracker. When the set target region is greater than 1.0× (e.g., ratios of 1.5 and 2.0), the target region will contain a large amount of background information, which makes it easy for the tracker to confuse the real target and background information in the template during the modeling process, and thus affects the tracking accuracy. From Table V, the tracker is able to obtain the best tracking results when N is set to 20. The proposed update module is to select the frame with the highest evaluation quality in the

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TABLE VI
TRACKING PERFORMANCE ABLATION

Method	UAV123-10fps		DTB70	
	Prec.	Succ.	Prec.	Succ.
Base	83.5	64.2	83.6	65.0
Base+TDM	86.3	66.0	85.3	66.1
Base+TDM+UPD-onlyD	85.8	65.9	84.9	66.5
Base+TDM+UPD (ETDMT)	86.9	66.6	85.8	67.1

update interval N. When N is set to small, the tracker can only select the optimal sample from a small number of samples to update, which has more interference than the optimal sample from a large number of samples, limiting the performance of the tracker. In addition, when N is set large, although the tracker is able to select better quality tracking samples, the tracker suffers from template obsolescence due to the large sampling interval, which also leads to the degradation of the tracker performance.

To verify the valid of the proposed template distinction modeling transformer network and the global–local spatial awareness update module, the trackers in the structure ablation experiment consist of Base, Base + TDM, Base + TDM + UPD-onlyD, and Base + TDM + UPD on the UAV123-10fps and DTB70 datasets. Base lacks both the template distinction modeling transformer network and the global–local spatial awareness update module. Base + TDM has the template distinction modeling transformer network but lacks the global–local spatial awareness update module. Base + TDM + UPD-onlyD includes the template distinction modeling transformer network and an update module that uses only the deepest feature of the network. Base + TDM + UPD is the proposed ETDMT, which includes both the template distinction modeling transformer network and a global–local spatial awareness update module.

From Table VI, it is evident that Base + TDM, with its template distinction modeling structure, significantly outperforms Base in both precision and success rate on the two datasets, showing improvements of 2.8% and 1.7% in precision on UAV123-10fps and DTB70, respectively. This indicates that template distinction modeling effectively improves the tracker's performance by suppressing the interaction between background and search areas in the template. Compared to Base + TDM, Base + TDM + UPD-onlyD, with an update module that uses only the deepest feature of the network, and focuses only on the local spatial information of the target leads to inaccurate updates and degradation of tracking performance. In contrast, Base + TDM + UPD with the global–local spatial awareness update module achieves improvements on both datasets. This validates the proposed update module's ability to accurately gather and utilize temporal context, providing the tracker with more timely discriminative information.

To further analyze the impact of each component on tracking performance, results from the UAV123-10fps attribute experiment are shown in Table VII. It is observed that Base + TDM, with template distinction modeling, significantly outperforms the conventional Base in background clutter, improving by 11.5% in precision and 9.0% in success rate.

Additionally, for FOC and POC attributes, Base + TDM also achieves significant improvements over Base. The template distinction modeling allows the tracker to accurately understand the real target, thus weakening the interference caused by occlusions. Compared to Base + TDM, Base + TDM + UPD shows comprehensive improvements in 11 tracking attributes by accurately learning the temporal context of the target, demonstrating the effectiveness of temporal context in enhancing tracking performance. In LR scenarios, the temporal context information enables the tracker to promptly learn the target's appearance under low illumination, resulting in a 2.2% increase in precision and a 1.7% success rate increase for Base + TDM + UPD-onlyD compared to Base + TDM. Base + TDM + UPD using multi-feature layer information for updates further improves precision and success rate by 0.5% in LR scenarios. It is worth noting that when Base + TDM + UPD has the update module, its performance in BC scenarios degrades. This is because the template distinction modeling transformer network divides the target and background areas according to the bounding box in the template, but in the BC scene, the bounding box in the updated template is prone to large deviations due to the background interference, which affects the accuracy of the target area division and reduces the tracking performance.

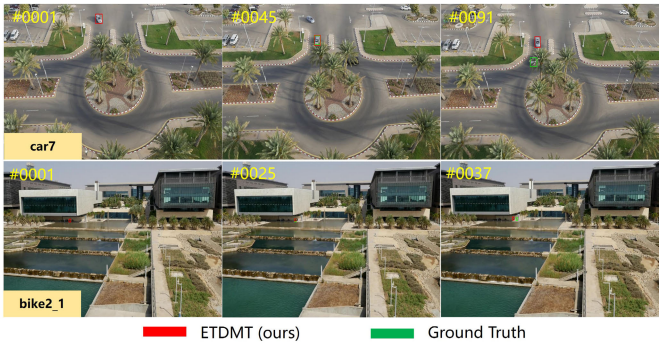


Fig. 6. Failure cases.

G. Failure Cases

Although the proposed tracker demonstrated competitive tracking performance in comparative experiments, it encountered tracking failures in scenarios with similar targets, as shown in Fig. 6. In the car7 sequence, the tracker initially tracks a white vehicle. However, in frame #45, the real target is heavily occluded, and there is an extremely similar white vehicle nearby, leading to erroneous tracking by the proposed tracker. In addition, when the occlusion ended, the tracker was still unable to reacquire the real target. In the bike2_1 sequence, the tracker follows a bicycle. At frame #25, the real target overlaps with a similar background object, followed by a rapid brightness change of the real target, causing the tracker to erroneously follow the similar background object and fail to recover the real target.

The reasons for these phenomena are analyzed. First, the proposed tracker lacks recapture capability, resulting in targets that reappear after occlusion often exceeding the tracker's search area. Moreover, the proposed tracker is based on image pair matching, not considering other discriminative information such as target trajectories. This leads to the possibility that when the appearance of the real target changes rapidly, the tracker may mistakenly believe that some background

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TABLE VII
ATTRIBUTE ABLATION

Method	ARC		BC		SV		SOB		CM		FOC		POC		OV		VC		LR		FM		IV	
	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.
Base	80.7	61.5	63.2	43.9	81.4	62.5	74.7	55.5	84.0	64.7	59.3	38.1	76.9	56.6	78.4	59.7	83.6	65.7	70.9	48.7	77.6	60.3	77.7	58.7
Base+TDM	84.5	64.0	74.7	52.9	84.6	64.5	85.2	63.5	88.9	67.9	71.9	47.9	81.2	59.8	81.8	62.4	88.6	69.0	72.0	48.6	77.3	58.7	80.4	60.8
Base+TDM+UPD-onlyD	82.9	63.1	65.7	46.3	84.0	64.2	86.1	63.7	88.0	67.5	69.6	45.5	80.9	59.7	82.3	63.4	87.8	68.4	74.2	50.3	79.8	60.8	78.9	59.4
Base+TDM+UPD	85.0	64.7	72.0	50.5	85.2	65.2	87.1	64.4	90.2	69.1	72.3	48.2	82.7	61.3	83.0	64.0	88.6	69.1	74.7	50.8	81.7	62.3	84.0	62.9

objects are more similar to the template. Therefore, it is necessary to explore a recapture method, which is used to recapture modules that have lost their targets after complete occlusion in a wider search area. In addition, more temporal information, such as target motion speed and direction, is also worth exploring for improving the aerial tracker's ability to discriminate between similar targets. Thus, the areas for further research are lost target recapture methods and efficient representation of more temporal information.

VI. CONCLUSION

In this article, we introduced the ETDMT tracker, which improves the robustness of the tracker in complex scenarios by constructing template distinction modeling and utilizing temporal contexts. The proposed template distinction modeling transformer network distinctly differentiates between the real target and background areas in the template, preventing the incorporation of background information from the template into the search area's modeling. This mitigates the interference of unavoidable background information in the template on target localization and discrimination. Furthermore, the global–local spatial awareness update module, designed to capture temporal context information, collects information from different semantic levels in the search area and fuses it to generate an updated score that assesses global environmental interference. This score is then combined with local target similarity scores to accurately learn the target's temporal context. Comparative experiments with a large number of trackers show that the proposed tracker achieves advanced tracking performance and efficient operation speeds. The results of ablation experiments prove the effectiveness of the proposed components. In the future, we plan to enhance the ETDMT tracker by adding a lightweight recapture module specifically for aerial tracking tasks and by incorporating a wider variety of discriminative information.

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